
Basic Concepts of String Theory

A 28-week seminar plan

Two semesters $\times$ 14 weeks $\times$ 2 hours

Design principle

The seminar should not attempt to reproduce every line of a 775-page graduate textbook on the board. Its job is to expose the logical spine, work the derivations that newcomers cannot reliably reconstruct alone, and teach participants how to recover the remaining details from the book.

Primary text	R. Blumenhagen, D. Lüst, and S. Theisen, <i>Basic Concepts of String Theory</i> (Springer, 2013)
Audience	Beginning graduate students with quantum mechanics, introductory QFT, special relativity, and some general relativity; prior string theory is not assumed
Meeting time	Two hours per week
Expected preparation	Varies strongly by background and purpose; see the four participant profiles in Sec. 1.2
Coverage strategy	Derive the structural results in full; discuss long model lists, appendices, and specialized examples at survey level
End point	A coherent path from the Polyakov string and worldsheet CFT to compactification, D-branes, fluxes, dualities, M-theory, F-theory, and AdS/CFT

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1 Pacing assumptions

Page-number convention

All page numbers in this plan are the *printed* page numbers in the book, not the PDF viewer's page count. The offset is not constant throughout the PDF, so use chapter and section titles as the definitive locator.

For a newcomer, this book is normally read at about five to eight serious pages per hour once equations are reconstructed. Thus a 25-page assignment is not a two-hour task: it is a four-to-six-hour preparation task. The seminar itself is where the group checks the conceptual map and two or three difficult steps.

1.1 Core versus survey coverage

Each week has two implicit levels:

- **Core** material should be understood well enough to reproduce the assumptions, derivation, and physical meaning. It receives board time.
- **Survey** material should be read for structure, examples, and notation. The speaker states the result and its role but does not derive every formula.

This distinction is what makes full-book coverage possible in 28 meetings. Without it, a careful seminar would need at least three semesters.

In every weekly road map below, the **teal bold node** is the result or derivation that should survive even if discussion runs slowly. The arrows show logical dependence, not a rigid timetable.

1.2 Expected time cost for different participants

The common cost is the two-hour meeting. The estimates below are additional weekly preparation time; advanced compactification weeks can require the upper end of each range.

Fresh participant intending to do research in string theory: 7–11 hours outside the seminar. Read all **Core** material and most **Survey** material, reconstruct the main derivation, and solve the weekly exercise. During BRST, superconformal CFT, Calabi–Yau compactification, and duality weeks, 10–13 hours is realistic. This participant should take speaker and checker roles regularly.

Fresh participant exploring the field: 3–5 hours outside the seminar. Read the chapter introduction, the **Core** subsections, and the closing discussion; follow one derivation carefully and treat the rest as orientation. Exercises are optional but strongly recommended in Weeks 2–9 and 11–15, where later material depends directly on the foundations.

Very fresh participant, including someone with limited QFT background: 2–4 hours for attendance, or 7–12 hours for technical progress. The lighter track consists of a conceptual preview and vocabulary sheet; it is enough to attend profitably but not to master the derivations. The technical track adds 3–6 hours of foundation repair in QFT, special relativity, path integrals, group theory, or differential geometry. Such a participant should not be expected to present BRST, modular invariance, or supergravity before completing the relevant bridge material.

Participant who has studied string theory once: 3–6 hours outside the seminar. Skim familiar sections, reconstruct the essential derivation from memory, and use a second reference to check conventions. If serving as speaker, add roughly 3–5 hours for preparing a coherent road map and anticipating questions from newcomers.

Role-dependent additions

Relative to the profile above, allow approximately 3–5 extra hours for the speaker, 1–2 hours for the checker, and 1–2 hours after the meeting for the scribe. These duties should rotate; otherwise the apparent workload of the seminar will be distributed very unevenly.

How to calibrate the pace with fresh students

At the end of Weeks 2, 5, and 8, anonymously ask: (i) preparation hours, (ii) fraction of the reading understood before the meeting, and (iii) which derivation remained opaque afterward. If the median preparation exceeds six hours, reduce the next assignment's **Survey** portion. Do not accelerate merely because the presenter is experienced.

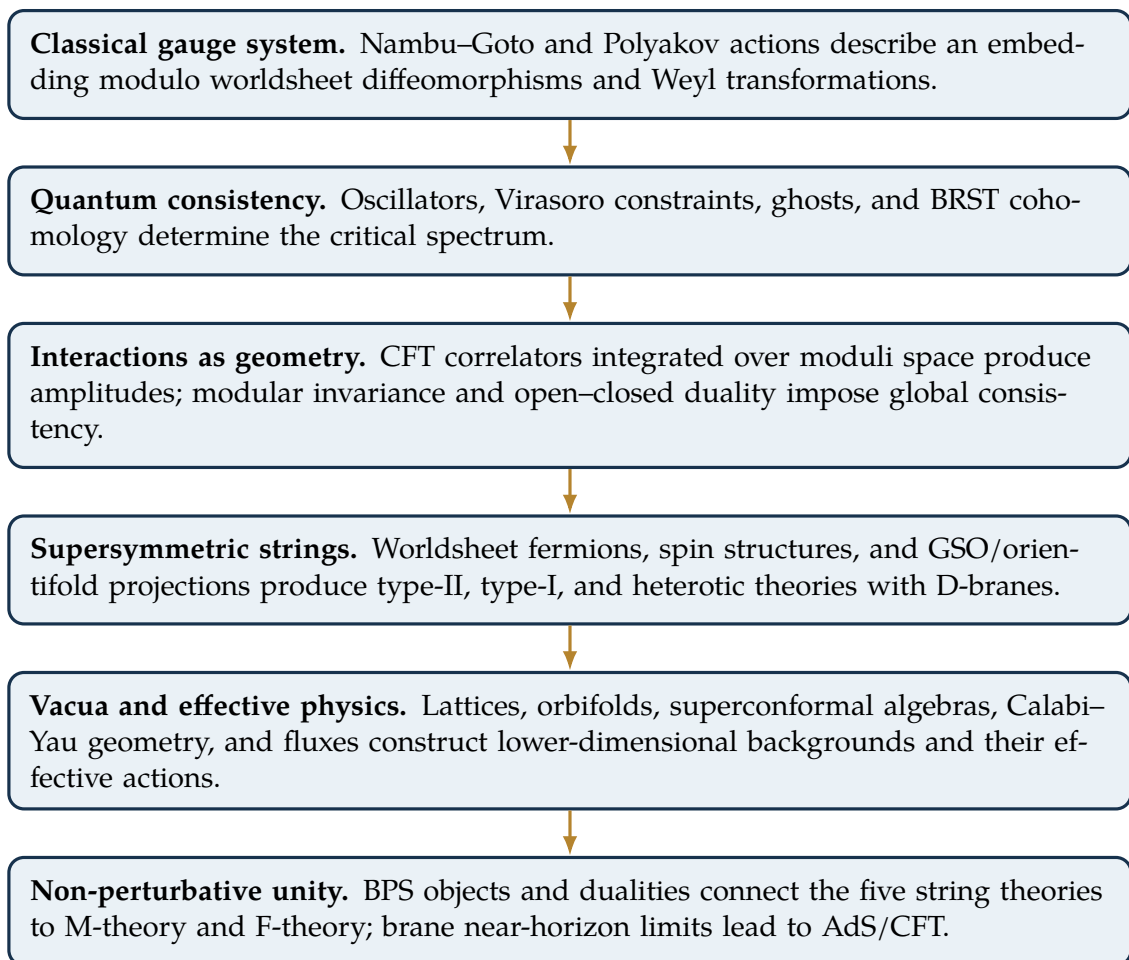
1.3 Roles for a participatory seminar

Use three rotating roles each week:

- the **speaker** prepares the logical map and the main derivation;
- the **checker** independently verifies conventions, signs, and one limiting case against a secondary source;
- the **scribe** produces a two-page note containing the final equations, unresolved questions, and corrections after discussion.

The speaker should circulate a one-page roadmap 24 hours before the meeting. Slides may be used for geometry, spectra, and summary tables, but the principal derivation should be done on a board or tablet slowly enough to invite questions.

2 The two-semester logic



Semester I establishes the first four boxes and ends with the consistent ten-dimensional superstrings. Semester II develops the last two boxes. The late chapters are not independent decorations: they repeatedly use BRST, modular invariance, boundary states, and supersymmetry from Semester I.

3 Semester I: Worldsheet foundations and ten-dimensional strings

Semester I objective

By Week 14, participants should be able to explain why the critical dimensions and spectra arise, how amplitudes are organized by worldsheet topology and moduli, and how spin structures and projections produce type-II and type-I superstrings with D-branes.

Week 1

Chs. 1; 2.1–2.3, pp. 1–22

Reading. Introduction; relativistic particle; Nambu–Goto and Polyakov actions and their symmetries. The particle review is **Survey** if the audience already knows reparametrization invariance.

Main idea. A relativistic string is a two-dimensional gauge system. The auxiliary worldsheet metric makes diffeomorphism and Weyl symmetry manifest, and its equation of motion imposes $T_{ab} = 0$.

Seminar road map. Motivation and course map $\rightarrow$ Nambu–Goto versus Polyakov $\rightarrow$ **worldsheet stress tensor, gauge symmetries, and $T_{ab} = 0$** $\rightarrow$ on-shell equivalence of the two actions.

Preparation and output. Beforehand, vary both actions and list all local/global symmetries. The scribe produces a convention sheet for signatures, α' , tension, and worldsheet coordinates.

Useful references. Tong, *Lectures on String Theory*, Sec. 1; Polchinski I, Secs. 1.2–1.3; Becker–Becker–Schwarz (BBS), Ch. 2.

Week 2

Ch. 2.4–2.5, pp. 23–34

Reading. Oscillator expansions and representative classical solutions.

Main idea. Boundary conditions select the mode expansion. The constraints remove longitudinal excitations and split closed-string motion into left- and right-movers tied by level matching.

Seminar road map. Boundary conditions $\rightarrow$ open/closed mode expansions $\rightarrow$ **Virasoro constraints and level matching** $\rightarrow$ center-of-mass motion, winding intuition, and representative classical solutions.

Preparation and output. Reconstruct one expansion without the book and check its canonical dimensions. The checker verifies signs for Neumann versus Dirichlet boundary terms.

Useful references. Tong, Sec. 1; Zwiebach, *A First Course in String Theory*, classical-string chapters; Polchinski I, Secs. 1.3–1.4.

Week 3

Ch. 3.1–3.2, pp. 35–45

Reading. Canonical and light-cone quantization of the bosonic string.

Main idea. Quantization converts modes into oscillators, while light-cone gauge solves the constraints explicitly and exposes the physical transverse Hilbert space.

Seminar road map. Canonical brackets $\rightarrow$ oscillator algebra $\rightarrow$ light-cone gauge $\rightarrow$ **physical transverse Hilbert space** $\rightarrow$ normal-ordering and Lorentz-anomaly preview.

Preparation and output. Compute the zero-point energy with a stated regulator. The speaker must distinguish a gauge choice, a constraint, and a physical-state condition.

Useful references. Tong, Sec. 2; Polchinski I, Ch. 2 and Sec. 4.1; BBS, Ch. 2.

Week 4

Ch. 3.3–3.5, pp. 46–62

Reading. Bosonic spectrum, covariant path-integral quantization, and Virasoro algebra.

Main idea. Critical dimension and intercept are quantum consistency conditions. The closed-string massless state already contains $G_{\mu\nu}$, $B_{\mu\nu}$, and Φ , so gravity is unavoidable.

Seminar road map. First open/closed mass levels $\rightarrow$ **critical dimension, intercept, and the closed massless fields** $\rightarrow$ Virasoro central extension $\rightarrow$ Lorentz/no-ghost consistency.

Preparation and output. Prepare a spectrum table for open and closed strings. The checker verifies state counts in $D = 26$ and identifies the tachyon's role and limitation.

Useful references. Tong, Sec. 2; Polchinski I, Ch. 4; BBS, Ch. 2.

Week 5

Ch. 4.1, pp. 63–84

Reading. Core two-dimensional CFT: conformal maps, OPEs, radial quantization, state–operator correspondence, Ward identities, and Virasoro symmetry.

Main idea. CFT turns local worldsheet dynamics into algebraic data. OPE coefficients and conformal weights control both states and amplitudes.

Seminar road map. Local conformal transformations $\rightarrow$ stress tensor and Ward identities $\rightarrow$ **TT OPE, Virasoro algebra, and state–operator map** $\rightarrow$ free-boson example.

Preparation and output. Every participant computes $T(z)X(w)$ and $T(z)e^{ikX(w)}$. The scribe begins a running table of fields, weights $(h, \bar{h})$, and central charges.

Useful references. Ginsparg, *Applied Conformal Field Theory*; Di Francesco–Mathieu–Sénéchal, Chs. 4–6; Tong, Sec. 4.

Week 6

Ch. 4.2–4.5, pp. 85–106

Reading. Closed-string application, boundary CFT, free-boson boundary states, and crosscap states.

Main idea. Closed strings use two chiral algebras; boundaries glue them and turn a spacetime brane into a worldsheet state. Crosscaps encode orientation reversal.

Seminar road map. Closed-string vertex conditions $\rightarrow$ conformal gluing $\rightarrow$ **Neumann/Dirichlet boundary states and their spacetime meaning** $\rightarrow$ Ishibashi/Cardy preview $\rightarrow$ crosscap conditions.

Preparation and output. Derive the oscillator gluing condition for one boundary state. Mark Cardy consistency as a target for Week 10 rather than treating it as a slogan now.

Useful references. Cardy, *Boundary Conformal Field Theory*; Polchinski I, Secs. 8.7–8.8; Recknagel–Schomerus, *Boundary Conformal Field Theory and the Worldsheet Approach to D-Branes*.

Week 7

Ch. 5, pp. 107–120

Reading. The bc ghost CFT and BRST quantization.

Main idea. Gauge fixing creates ghosts whose central charge cancels that of matter in the critical theory. BRST cohomology defines physical states covariantly and identifies null states with gauge redundancy.

Seminar road map. bc ghost system $\rightarrow$ ghost stress tensor and $c_{gh} = -26$ $\rightarrow$ BRST charge $\rightarrow$ **nilpotency and physical-state cohomology** $\rightarrow$ massless gauge redundancy.

Preparation and output. Show explicitly that a BRST-exact massless polarization is a linearized gauge transformation. The checker tracks ghost number and zero modes.

Useful references. Polchinski I, Chs. 2–4; Tong, Sec. 5; BBS, Secs. 3.1–3.4.

Week 8

Ch. 6.1–6.2, pp. 121–145

Reading. Genus expansion and the Polyakov path integral for closed bosonic strings.

Main idea. Interactions are sums over worldsheet topologies and integrals over moduli, not a local spacetime interaction vertex. Conformal Killing symmetries explain fixed vertex positions and ghost insertions.

Seminar road map. Worldsheet topology and powers of g_s $\rightarrow$ metric gauge fixing $\rightarrow$ **moduli, conformal Killing symmetries, and ghost insertions** $\rightarrow$ sphere amplitude and the general measure.

Preparation and output. Prepare the three-tachyon sphere amplitude and identify every volume divided out. Use Tong Problem Sheet 3 for practice.

Useful references. Tong, Sec. 6 and Problem Sheet 3; Polchinski I, Chs. 5–6; BBS, Ch. 3.

Week 9

Ch. 6.3–6.4, pp. 146–156

Reading. Torus partition function and modular invariance; torus partition functions of rational CFTs.

Main idea. One-loop consistency is invariance under large worldsheet diffeomorphisms. Characters furnish a finite-dimensional modular representation, and modular invariants pair left and right sectors consistently.

Seminar road map. Free-boson torus partition function $\rightarrow$ fundamental domain $\rightarrow$ **modular S, T transformations and modular invariance** $\rightarrow$ RCFT characters and $[M, S] = [M, T] = 0$.

Preparation and output. Each participant checks one modular transformation using $\eta(\tau)$ or theta functions. The scribe records the distinction among modular covariance, invariance, and anomaly.

Useful references. Ginsparg, torus/free-field sections; Polchinski I, Ch. 7; Di Francesco–Mathieu–Sénéchal, modular-invariance chapters.

Week 10

Ch. 6.5–6.8, pp. 157–174

Reading. Cylinder amplitudes, RCFT boundary states, Klein bottle and Möbius strip, and D-brane tension.

Main idea. Worldsheet duality identifies an open-string loop with closed-string exchange. Cardy and tadpole consistency translate modular transformations into restrictions on boundary conditions, spectra, and charges.

Seminar road map. Cylinder in the open channel $\rightarrow$ modular transformation $\rightarrow$ closed-string exchange $\rightarrow$ **Cardy and tadpole consistency** $\rightarrow$ Klein bottle/Möbius amplitudes and D-brane tension.

Preparation and output. Draw all one-loop surfaces and label their direct/transverse channels. The checker follows the normalization of one boundary state.

Useful references. Cardy, *Boundary Conformal Field Theory*; Johnson, *D-Brane Primer*; Polchinski I, Ch. 8.

Week 11

Ch. 7, pp. 175–194

Reading. Classical RNS string: motivation, locally supersymmetric action, superconformal gauge, NS/R boundary conditions, and mode expansions.

Main idea. Worldsheet supersymmetry pairs X^μ with ψ^μ . Periodic and antiperiodic fermions create Ramond and Neveu–Schwarz sectors with different zero-mode structures.

Seminar road map. RNS action and local symmetries $\rightarrow$ superconformal gauge $\rightarrow$ NS/R boundary conditions and mode expansions $\rightarrow$ **Ramond zero modes as a Clifford algebra and the meaning of the two sectors.**

Preparation and output. Prepare a boson/fermion comparison table. Treat the two-dimensional spinor appendix as reference unless conventions are causing difficulty.

Useful references. BBS, Ch. 4; Green–Schwarz–Witten (GSW) I, Chs. 3–4; Polchinski II, Ch. 10.

Week 12

Ch. 8, pp. 195–222

Reading. Quantized fermionic string: super-Virasoro algebra, light-cone spectrum, GSO projection, path integral, and spacetime spinors.

Main idea. Quantum consistency fixes $D = 10$. The GSO projection removes the tachyon and creates spacetime supersymmetry by correlating fermion number, chirality, and modular consistency.

Seminar road map. NS/R oscillator quantization $\rightarrow$ normal-ordering constants and first levels $\rightarrow D = 10$, **GSO projection, and the projected spectrum** $\rightarrow$ path-integral/superghost overview.

Preparation and output. Construct the projected NS and R ground states and count on-shell degrees of freedom. The checker verifies ten-dimensional chirality conventions.

Useful references. BBS, Chs. 4–5; GSW I, Chs. 4–5; Polchinski II, Ch. 10.

Week 13

Ch. 9.1–9.3, pp. 223–240

Reading. Spin structures and superstring partition functions; fermionic boundary states; type-II D-branes.

Main idea. The spin-structure sum realizes modular invariance and spacetime supersymmetry. Fermionic boundary states show that type-II D-branes preserve half the supercharges and source R–R fields.

Seminar road map. Spin structures and theta functions $\rightarrow$ type-II torus sum $\rightarrow$ fermionic gluing $\rightarrow$ **GSO-projected boundary states, preserved supersymmetry, and R–R charge** $\rightarrow$ BPS interpretation.

Preparation and output. Verify one theta-function transformation and derive the supersymmetry projector for a Dp -brane. The detailed theta appendix may be consulted as needed.

Useful references. Polchinski II, Chs. 10 and 13; BBS, Chs. 5–6; Johnson, *D-Brane Primer*.

Week 14

Ch. 9.4–9.6, pp. 241–262

Reading. Type-I string, stable non-BPS branes, theta-function appendix, and Semester I synthesis.

Main idea. Orientifolding relates oriented and unoriented theories. Cancellation of massless tadpoles fixes the type-I Chan–Paton group, while K-theoretic/stability ideas foreshadow non-perturbative brane physics.

Seminar road map. Worldsheet parity $\rightarrow$ type-I spectrum $\rightarrow$ **annulus–Möbius–Klein-bottle tadpole cancellation and $SO(32)$** $\rightarrow$ one non-BPS brane example $\rightarrow$ Semester I synthesis.

Preparation and output. Prepare a one-page chain from Polyakov symmetry to type-I consistency. Use the final synthesis block as a recovery buffer if the semester is one meeting behind; do not omit the tadpole argument.

Useful references. Polchinski II, Chs. 10 and 13; BBS, Chs. 5–6; Johnson, *D-Brane Primer*.

4 Semester II: Compactification, effective theory, and duality

Semester II objective

By Week 28, participants should be able to move among worldsheet CFT, target-space geometry, and low-energy effective theory; explain how fluxes and branes modify compactification; and use BPS charges and duality to connect string theory, M-theory, F-theory, and holography.

Week 15

Ch. 10.1–10.3, pp. 263–283

Reading. Toroidal compactification of the closed bosonic string and toroidal partition functions.
Main idea. Momentum and winding form an even self-dual lattice. T-duality is an exact equivalence of CFTs, not merely a low-energy symmetry, and exchanges geometry with stringy quantum numbers.

Seminar road map. Circle momentum and winding $\rightarrow$ level matching $\rightarrow R \leftrightarrow \alpha'/R$, **Narain lattice, and modular invariance** $\rightarrow$ enhanced symmetry at the self-dual radius.

Preparation and output. Compute the mass spectrum under T-duality and identify the extra massless states at the self-dual radius. The checker verifies lattice signature and normalization.

Useful references. Polchinski I, Ch. 8; Giveon–Porrati–Rabinovici, *Target Space Duality in String Theory*; Tong, Sec. 8.

Week 16

Ch. 10.4–10.6, pp. 284–320; preview 15.2–15.3

Reading. Ten-dimensional heterotic strings, toroidal orbifolds, compact D-branes, and a preview of supersymmetric orbifolds/orientifolds.

Main idea. The heterotic string uses asymmetric left/right movers; modular invariance selects $E_8 \times E_8$ and $SO(32)$. Orbifolding combines a quotient with twisted sectors, while D-branes and orientifolds add localized sectors and tadpole constraints.

Seminar road map. Asymmetric heterotic construction $\rightarrow$ **even self-dual gauge lattices, modular invariance, and the two gauge groups** $\rightarrow$ orbifold untwisted/twisted sectors $\rightarrow$ compact branes and images.

Preparation and output. Prepare an untwisted/twisted-sector table. Read Ch. 15.2–15.3 only for the four-dimensional supersymmetric destination; details return in Week 23.

Useful references. Polchinski II, Chs. 11 and 16; BBS, Ch. 7; Ginsparg, orbifold sections.

Week 17

Ch. 11.1–11.4, pp. 321–342

Reading. Kač–Moody algebras, root/weight lattices, Frenkel–Kač–Segal construction, and bosonization.

Main idea. Affine currents encode non-Abelian symmetry on the worldsheet. Free bosons on special lattices and free fermions provide equivalent realizations of current algebras.

Seminar road map. Current OPE and affine commutators $\rightarrow$ **Sugawara stress tensor and central charge** $\rightarrow SU(2)$ lattice/FKS construction $\rightarrow$ fermionic realization and bosonization.

Preparation and output. Everyone computes the conformal weight of a current and one bosonized operator. Keep general Lie-algebra tables at **Survey** level.

Useful references. Ginsparg, affine-current and bosonization sections; Di Francesco–Mathieu–Sénéchal, affine Lie algebra chapters; Polchinski II, Secs. 15.2–15.5.

Week 18

Ch. 11.5–11.6; 12.1, pp. 343–369

Reading. Affine characters and integrable representations, the $\widehat{su}(2)_k$ example, and $\mathcal{N} = 1$ superconformal symmetry.

Main idea. Unitarity truncates affine representations at fixed level, producing the finite character data used by RCFT. Adding a supercurrent enlarges Virasoro symmetry and organizes NS/R representations.

Seminar road map. Integrable highest-weight representations $\rightarrow$ affine characters and modular data $\rightarrow \mathcal{N} = 1$ **super-Virasoro algebra** $\rightarrow$ NS/R representation structure.

Preparation and output. Work the $k = 1$ or $k = 2$ $SU(2)$ example and check one modular transformation. The scribe links the result to Weeks 9 and 12.

Useful references. Di Francesco–Mathieu–Sénéchal, affine and superconformal chapters; Ginsparg; Polchinski II, Ch. 15.

Week 19

Ch. 12.2–12.3, pp. 370–390; preview 15.4–15.5

Reading. $\mathcal{N} = 2$ superconformal symmetry, chiral rings, topological twisting, and the worldsheet logic behind Calabi–Yau/Gepner vacua.

Main idea. The $\mathcal{N} = 2$ algebra identifies protected chiral primaries. Spectral flow connects NS and R sectors, while twisting turns a supercharge into a BRST operator and isolates topological data.

Seminar road map. $\mathcal{N} = 2$ algebra $\rightarrow$ **chiral-primary bound and spectral flow** $\rightarrow$ chiral ring $\rightarrow$ A/B topological twists $\rightarrow$ Gepner-model preview.

Preparation and output. Prepare one minimal-model chiral ring or sigma-model example. The checker distinguishes worldsheet BRST from topological BRST.

Useful references. Hori et al., *Mirror Symmetry*, relevant SCFT chapters; Polchinski II, Sec. 19.1; Di Francesco–Mathieu–Sénéchal, $\mathcal{N} = 2$ chapters.

Week 20

Ch. 13, pp. 391–426; connect 15.6

Reading. Bosonization and first-order systems; covariant RNS vertex operators; BRST and picture changing; D-brane supersymmetry; covariant lattices and heterotic models.

Main idea. Covariant vertices translate the light-cone spectrum into calculable operators. Ghost pictures and GSO/cocycle data are bookkeeping for a consistent operator algebra, while covariant lattices package spacetime and gauge quantum numbers.

Seminar road map. $\beta\gamma$ bosonization and picture number $\rightarrow$ NS/R vertex operators $\rightarrow$ **BRST conditions and picture-changing equivalence** $\rightarrow$ spacetime supercharges/D-brane projector $\rightarrow$ covariant lattices.

Preparation and output. Each participant checks conformal weight, picture, ghost number, and BRST condition of one vertex. Treat long lattice classifications as **Survey**.

Useful references. Polchinski II, Chs. 10–12; Friedan–Martinec–Shenker formalism as historical background; BBS, Ch. 5.

Week 21

Ch. 14.1–14.4, pp. 427–472

Reading. Worldsheet conformal invariance and spacetime geometry; Buscher rules; compactification logic; essential differential and complex geometry.

Main idea. A nonlinear sigma model makes target-space fields into worldsheet couplings. Vanishing beta functions become spacetime field equations, and gauging an isometry on the worldsheet derives T-duality.

Seminar road map. Metric/ B/Φ sigma model $\rightarrow$ beta functions and spacetime equations $\rightarrow$ **gauged-isometry derivation of the Buscher rules** $\rightarrow$ compactification logic and geometry dictionary.

Preparation and output. Reconstruct the one-loop metric beta function schematically and track the dilaton shift under T-duality. Mathematical preliminaries are reference, not a second geometry course.

Useful references. Tong, Secs. 7–8; Giveon–Porrati–Rabinovici; Hori et al., geometry chapters.

Week 22

Ch. 14.5–14.8, pp. 473–520

Reading. Calabi–Yau manifolds; type-II and heterotic compactification on CY threefolds; Riemannian/Hodge appendix.

Main idea. Calabi–Yau holonomy converts geometry into preserved supersymmetry. Hodge numbers count moduli and multiplets, while the heterotic bundle must satisfy holomorphy, stability/supersymmetry, and the Bianchi identity.

Seminar road map. Covariantly constant spinor and $SU(3)$ holonomy $\rightarrow$ Hodge decomposition $\rightarrow$ **IIA/IIB moduli and multiplet counting** $\rightarrow$ heterotic bundles, Bianchi identity, and standard embedding.

Preparation and output. Prepare the IIA/IIB Hodge-number dictionary and one heterotic anomaly/Bianchi identity check. Detailed curvature identities remain **Survey**.

Useful references. Hori et al., *Mirror Symmetry*; Polchinski II, Chs. 17 and 19; BBS, Ch. 9.

Week 23

Ch. 15, pp. 521–584, integrated review

Reading. CFTs for type-II and heterotic vacua: supersymmetric orbifolds, orientifolds, worldsheet supersymmetry, Gepner models, and covariant-lattice heterotic vacua.

Main idea. Exact CFT gives access to string-scale or non-geometric points where a large-radius manifold description is insufficient. Orbifold, orientifold, Gepner, and lattice constructions are different implementations of the same consistency requirements.

Seminar road map. Common construction data $\rightarrow$ supersymmetric orbifold example $\rightarrow$ **Gepner tensor product, GSO projection, and shared consistency conditions** $\rightarrow$ heterotic covariant-lattice example.

Preparation and output. This is a synthesis week: Sections 15.2–15.6 were previewed in Weeks 16, 19, and 20. Each group presents one model with columns “CFT data, projection, spectrum, moduli, consistency checks.”

Useful references. Polchinski II, Chs. 16 and 19; Hori et al., SCFT and mirror-symmetry chapters; Ginsparg, orbifold/current-algebra sections.

Week 24

Ch. 16.1–16.3, pp. 585–621

Reading. String scattering amplitudes and extraction of the low-energy effective field theory.

Main idea. Vertex-operator correlators produce an analytic S -matrix whose poles encode the entire string spectrum. The α' expansion separates field-theory exchange from local higher-derivative corrections.

Seminar road map. Vertex-operator correlator and ghost pictures $\rightarrow$ Veneziano or Virasoro-Shapiro structure $\rightarrow$ **poles, residues, and factorization** $\rightarrow$ low-energy α' expansion and operator matching.

Preparation and output. Every participant expands one Gamma-function ratio and identifies exchange versus contact terms. Use the appendix integrals only as needed.

Useful references. Tong, Sec. 6 and Problem Sheet 3; Polchinski I, Chs. 5–7; BBS, Ch. 3.

Week 25

Ch. 16.4–16.6, pp. 622–640

Reading. Ten- and eleven-dimensional supergravities, D-brane effective actions, and amplitude integrals.

Main idea. Low-energy supergravity is fixed by massless string fields and supersymmetry; D-branes carry DBI and Wess–Zumino actions whose tensions and charges are matched by amplitudes and duality.

Seminar road map. Type-II/heterotic/11D massless fields $\rightarrow$ string versus Einstein frame $\rightarrow$ **DBI plus Wess–Zumino actions** $\rightarrow$ R–R charge, tension, and low-field limits.

Preparation and output. Prepare a frame/coupling convention sheet and expand DBI to recover Maxwell theory. The checker verifies dimensions and powers of g_s and α' .

Useful references. Tong, Sec. 7; Johnson, *D-Brane Primer*; BBS, Chs. 6 and 8.

Week 26

Ch. 17, pp. 641–674

Reading. Type-II compactifications with D-branes and fluxes: supersymmetry, tadpoles, massless spectra, flux compactification, and $SU(3)$ structure.

Main idea. Branes, orientifolds, and fluxes generate charges, superpotentials, and moduli potentials. Supersymmetry is encoded by spinor/structure conditions, while integrated Bianchi identities impose global tadpole constraints.

Seminar road map. Orientifold/brane setup $\rightarrow$ tadpole/Bianchi constraints $\rightarrow$ **flux superpotential and supersymmetry conditions** $\rightarrow$ moduli stabilization $\rightarrow$ $SU(3)$ structure and torsion-class overview.

Preparation and output. Work one simple flux/tadpole counting example. Keep detailed model spectra and the appendix at **Survey** level unless the seminar specializes in compactification.

Useful references. Graña, *Flux Compactifications in String Theory*; Douglas–Kachru, *Flux Compactification*; BBS, Ch. 10.

Week 27

Ch. 18.1–18.7, pp. 675–723

Reading. Extended objects, central charges and BPS bounds, supergravity branes, non-perturbative dualities, and M-theory.

Main idea. BPS quantities can be followed from weak to strong coupling. Their protected charges and tensions show that the five ten-dimensional string theories are limits of one network whose eleven-dimensional corner is M-theory.

Seminar road map. Supersymmetry algebra and BPS bound $\rightarrow$ brane charge and supergravity solution $\rightarrow$ **tension scaling and the S/T -duality network** $\rightarrow$ IIA strong coupling and $R_{11} = g_s \ell_s$.

Preparation and output. Build a duality table for F1, Dp , NS5, KK modes, M2, and M5. The checker tests dimensional reduction and coupling powers rather than memorizing arrows.

Useful references. BBS, Ch. 8; Polchinski II, Ch. 14; Johnson, *D-Brane Primer*; Giveon–Porrati–Rabinovici.

Week 28

Ch. 18.8–18.9, pp. 724–774; global synthesis

Reading. F-theory, AdS/CFT, and the complete logical map of the book.

Main idea. F-theory geometrizes the varying type-IIB axio-dilaton using an elliptic fibration. AdS/CFT arises when a brane decoupling limit has both a worldvolume QFT description and a near-horizon gravitational description.

Seminar road map. 7-brane monodromy $\rightarrow$ F-theory elliptic-fibration dictionary $\rightarrow$ D3-brane near-horizon limit $\rightarrow$ **AdS/CFT decoupling argument and parameter map** $\rightarrow$ global synthesis and open questions.

Preparation and output. Prepare a short explanation of one arrow in the full course map. The final written product is a shared “twenty indispensable equations” document with assumptions and domains of validity.

Useful references. Weigand, *TASI Lectures on F-theory*; Maldacena, *The Large N Limit of Superconformal Field Theories and Supergravity*; Aharony et al., *Large N Field Theories, String Theory and Gravity*.

5 Reference library and how to use it

5.1 Free or openly accessible anchors

- David Tong, *Lectures on String Theory*. The most useful alternate first pass for Weeks 1–10 and 21–25. The accompanying problem sheets cover classical and quantum strings, CFT, scattering amplitudes, and background fields.
- Paul Ginsparg, *Applied Conformal Field Theory*. Free CFT support for Weeks 5, 9, and 16–18, especially free fields, torus partition functions, current algebras, and orbifolds.
- John Cardy, *Boundary Conformal Field Theory*. A compact conceptual supplement for Weeks 6 and 10, with a clean route from conformal boundaries to Cardy consistency.
- Clifford Johnson, *D-Brane Primer*. Useful for boundary conditions, charges, effective actions, and dualities in Weeks 10, 13–14, and 25–27.
- Giveon, Porrati, and Rabinovici, *Target Space Duality in String Theory*. A systematic T-duality and Narain-lattice reference for Weeks 15, 21, and 27.
- Hori et al., *Mirror Symmetry*. The principal bridge between geometry and $\mathcal{N} = 2$ SCFT for Weeks 19 and 21–23. Assign selected chapters only.

- Mariana Graña, *Flux Compactifications in String Theory*. Detailed backup for Week 26; the introductory and supersymmetry sections are the most suitable for a general seminar.
- Timo Weigand, *TASI Lectures on F-theory*. A modern geometric supplement for Week 28. Begin with the F-theory dictionary and codimension-one physics.
- Aharony et al., *Large N Field Theories, String Theory and Gravity*. A broad AdS/CFT review for Week 28; the introductory dictionary and D3-brane decoupling limit are the relevant starting points.

5.2 Books for checking derivations

- J. Polchinski, *String Theory*, Vol. I, is the main technical cross-check for bosonic strings, CFT, path integrals, amplitudes, and toroidal compactification (Weeks 1–10 and 15).
- J. Polchinski, *String Theory*, Vol. II, is the main cross-check for superstrings, D-branes, duality, orbifolds, and Calabi–Yau compactification (Weeks 11–23 and 27).
- K. Becker, M. Becker, and J. Schwarz, *String Theory and M-Theory: A Modern Introduction*, gives a complementary target-space presentation, especially useful in Weeks 21–28.
- P. Di Francesco, P. Mathieu, and D. Sénéchal, *Conformal Field Theory*, is the detailed algebraic reference for Weeks 5–6, 9, and 17–19.
- M. Green, J. Schwarz, and E. Witten, *Superstring Theory*, Vol. I, is useful when Weeks 11–13 require a second derivation in RNS conventions.

Do not assign every reference

For each week, the book plus *one* secondary source is enough. The speaker uses a second technical source to check the derivation; newcomers should not be asked to reconcile three convention systems simultaneously.

6 Exercise menu for every seminar

Exercise numbers vary between editions and printings, so the assignments below identify a source chapter or problem sheet and the mathematical skill to be practised. The organizer can then select the exact numbered problem available to the group. “Adapt” means that the speaker should cover the source’s solution and restate one intermediate calculation as a self-contained problem. For brevity, BBS denotes Blumenhagen–Brinkmann–Schlechter.

6.1 Semester I: worldsheet foundations and ten-dimensional strings

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| Week 1 | Tong, Problem Sheet 1, or Polchinski I, Ch. 1: vary the Polyakov action, derive T_{ab} , and show why its vanishing is a constraint rather than an ordinary equation of motion. |
| Week 2 | Tong, Problem Sheet 1, or Polchinski I, Ch. 1: obtain open- and closed-string mode expansions from the boundary conditions and derive the classical Virasoro generators. |
| Week 3 | Tong, Problem Sheet 1, or Polchinski I, Chs. 2 and 4: quantize the oscillators, compute the normal-ordering contribution, and check the physical-state constraints at the first two levels. |
| Week 4 | Polchinski I, Ch. 4, or BBS, Ch. 2: count physical oscillator states and decompose the closed-string massless state into graviton, two-form, and dilaton polarizations. |

- Week 5** Tong, Problem Sheet 2, or Di Francesco–Mathieu–Sénéchal, Chs. 5–6: derive the free-boson OPE, the conformal Ward identity, and the weight of $e^{ik \cdot X}$.
- Week 6** Polchinski I, Ch. 8, or Cardy’s boundary-CFT exercises: impose a boundary gluing condition, construct the corresponding oscillator overlap, and interpret the result as a boundary state.
- Week 7** Polchinski I, Chs. 3–4, or an adapted Tong CFT problem: compute the bc central charge, identify the condition for $Q_B^2 = 0$, and distinguish BRST-exact from BRST-closed states in a simple example.
- Week 8** Tong, Problem Sheet 3, or Polchinski I, Ch. 5: fix three points on the sphere, evaluate the conformal-Killing-group Jacobian, and reproduce a basic ghost correlator.
- Week 9** Polchinski I, Ch. 7, or Ginsparg’s CFT notes: derive the modular transformation of $\eta(\tau)$ or a theta function and verify modular invariance of the compact free-boson partition function.
- Week 10** Polchinski I, Chs. 7–8, or Cardy’s boundary-CFT exercises: transform a cylinder amplitude between open and closed channels and extract one Cardy-consistency condition.
- Week 11** BBS, Ch. 4; Green–Schwarz–Witten I, Chs. 3–4; or Polchinski II, Ch. 10: derive NS and R mode expansions and build the Ramond zero-mode Clifford algebra.
- Week 12** Polchinski II, Ch. 10, or BBS, Chs. 4–5: determine the NS normal-ordering constant, apply the GSO projection, and count the first surviving physical states.
- Week 13** Polchinski II, Chs. 10 and 13, or BBS, Chs. 5–6: prove one theta-function cancellation or construct one orientifold/boundary projector and read off its spectrum.
- Week 14** Polchinski II, Chs. 10 and 13, or Johnson’s *D-Branes*: write the annulus, Möbius-strip, and Klein-bottle contributions in both channels and use tadpole cancellation to recover the $SO(32)$ condition.

6.2 Semester II: duality, compactification, branes, and M-theory

- Week 15** Polchinski I, Ch. 8, or Tong, Problem Sheet 4: derive the momentum–winding spectrum, prove its T-duality invariance, and identify the extra massless states at the self-dual radius.
- Week 16** Polchinski II, Chs. 11 and 16, or BBS, Ch. 7: check evenness and self-duality for a simple lattice, or compute the untwisted and twisted sectors of a basic orbifold.
- Week 17** Di Francesco–Mathieu–Sénéchal’s affine-algebra exercises, Ginsparg’s notes, or Polchinski II, Ch. 15: derive the Sugawara central charge and work out the characters of a simple model such as $SU(2)_1$.
- Week 18** Di Francesco–Mathieu–Sénéchal or Polchinski II, Ch. 15: determine the integrable weights at fixed level, verify one character transformation, and derive one $\mathcal{N} = 1$ superconformal mode relation.
- Week 19** Hori et al., *Mirror Symmetry*, or Polchinski II, Ch. 19: prove the chiral-primary bound, apply spectral flow, and multiply two simple chiral-ring elements.
- Week 20** Polchinski II, Chs. 10–12: check the conformal weights and ghost pictures of NS and R vertex operators, test BRST closure, and relate two pictures with a picture-changing operator.

- Week 21** Tong, Problem Sheet 4; Polchinski I, Ch. 8; or Hori et al.: compute a one-loop sigma-model beta function or carry out the Buscher dualization for a background with one Abelian isometry.
- Week 22** Polchinski II, Chs. 17 and 19, or Hori et al.: use Hodge data to count a Calabi–Yau moduli sector, or verify the cohomological consistency of the heterotic Bianchi identity.
- Week 23** Polchinski II, Chs. 16 and 19, or Hori et al.: compute a simple orbifold projection, a Gepner-model charge selection rule, or the corresponding spacetime GSO condition.
- Week 24** Tong, Problem Sheet 3, or Polchinski I, Chs. 6–7: analyze the poles and residues of a four-point amplitude and identify the exchanged string states or its low-energy field-theory limit.
- Week 25** Tong, Problem Sheet 4; Polchinski II, Ch. 13; or BBS, Ch. 6: expand the DBI action to quadratic order and match its gauge coupling or scalar kinetic terms to open-string data.
- Week 26** BBS, Ch. 10, or an adapted exercise from the flux review by Graña: evaluate a simple flux superpotential, tadpole, or $SU(3)$ -structure condition.
- Week 27** Polchinski II, Ch. 14, or BBS, Ch. 8: compare BPS brane tensions under S-duality or derive the relation between the M-theory circle radius and the type-IIA coupling.
- Week 28** BBS, Ch. 12, plus an adapted problem from Weigand’s F-theory notes or the AdS/CFT review by Aharony et al.: compute one seven-brane monodromy or one AdS/CFT parameter map.

A sustainable assignment rule

For each seminar choose one core problem from the menu. Mark one further part as an optional extension, and permit exploratory listeners to submit only a solution outline. At the next meeting, discuss the method and one diagnostic mistake; a complete blackboard solution is needed only when it supplies a tool used later.

7 Contingency rules

If the group falls behind

One week: keep the BRST and RNS foundations intact. Compress Week 14’s model catalogue, Ch. 10.6, and non-BPS examples; make Week 23 a comparison of one representative construction. **Two weeks:** preserve Weeks 1–9, 11–13, 15, 19–22, and 24–28. In Weeks 10, 14, 16, 18, and 23, derive one model correctly and treat the remaining examples as survey material.

What successful completion means

Success means being able to choose the appropriate description, state its consistency condition, reconstruct the central derivation, and locate the detailed formula—not memorizing every lattice or supergravity coefficient.